

Modular Groups Exercises

Q&A Notes

Question 1

Draw the group operation tables for $(\mathbb{Z}_4, +_4)$ and $(\mathbb{Z}_5^*, \cdot_5)$. Show that these groups are isomorphic by giving a one-to-one correspondence α such that

$$a + b \equiv c \pmod{4}$$

if and only if

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

Answer 1

The two groups are isomorphic. The key correspondence is

$$\alpha(0) = 1, \quad \alpha(1) = 2, \quad \alpha(2) = 4, \quad \alpha(3) = 3.$$

Step 1: Build the table for $(\mathbb{Z}_4, +_4)$

Here

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}.$$

The operation is: add normally, then take the remainder modulo 4.

For example,

$$2 + 3 = 5 \equiv 1 \pmod{4}.$$

So the entry in row 2, column 3, is 1.

The full operation table is:

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Step 2: Build the table for $(\mathbb{Z}_5^*, \cdot_5)$

By definition,

$$\mathbb{Z}_5^* = \{a : 1 \leq a < 5, \gcd(a, 5) = 1\}.$$

Since 5 is prime, every nonzero residue modulo 5 is relatively prime to 5. Therefore

$$\mathbb{Z}_5^* = \{1, 2, 3, 4\},$$

The operation is: multiply normally, then take the remainder modulo 5.

For example,

$$3 \cdot 4 = 12 \equiv 2 \pmod{5}.$$

So the entry in row 3, column 4, is 2.

The full operation table is:

$\cdot_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Step 3: Choose the matching

We need a one-to-one map

$$\alpha : \mathbb{Z}_4 \rightarrow \mathbb{Z}_5^*.$$

The useful matching comes from powers of 2 modulo 5:

$$2^0 \equiv 1, \quad 2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 3 \pmod{5}.$$

So

$$\boxed{\alpha(k) = 2^k \pmod{5}.}$$

This gives

$$0 \mapsto 1, \quad 1 \mapsto 2, \quad 2 \mapsto 4, \quad 3 \mapsto 3.$$

Step 4: Check that addition becomes multiplication

We must show that

$$a + b \equiv c \pmod{4}$$

if and only if

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

Since

$$\alpha(a) = 2^a \quad \text{and} \quad \alpha(b) = 2^b,$$

Then

$$\alpha(a)\alpha(b) = 2^a 2^b = 2^{a+b}.$$

If

$$a + b \equiv c \pmod{4},$$

then $a + b = c + 4t$ for some integer t . Hence

$$2^{a+b} = 2^{c+4t} = 2^c (2^4)^t.$$

But

$$2^4 = 16 \equiv 1 \pmod{5}.$$

Therefore

$$2^{a+b} \equiv 2^c \pmod{5}.$$

So

$$\alpha(a)\alpha(b) \equiv \alpha(c) \pmod{5}.$$

Concrete check

Take

$$a = 2, \quad b = 3.$$

In $\mathbb{Z}_4$,

$$2 + 3 = 5 \equiv 1 \pmod{4}.$$

So $c = 1$.

Apply α :

$$\alpha(2) = 4, \quad \alpha(3) = 3, \quad \alpha(1) = 2.$$

Now check multiplication modulo 5:

$$\alpha(2)\alpha(3) = 4 \cdot 3 = 12 \equiv 2 \pmod{5}.$$

Since

$$\alpha(1) = 2,$$

we get

$$\alpha(2)\alpha(3) \equiv \alpha(1) \pmod{5}.$$

Therefore,

$$\boxed{(\mathbb{Z}_4, +_4) \cong (\mathbb{Z}_5^*, \cdot_5)}.$$

Question 2

List all subgroups of $\mathbb{Z}_9$ and of $\mathbb{Z}_{13}^*$.

Answer 2

Part A: Subgroups of $\mathbb{Z}_9$

The group is

$$\mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\},$$

with addition modulo 9.

A subgroup of $\mathbb{Z}_9$ must contain the identity 0, must be closed under addition modulo 9, and its size must divide the group size.

Since

$$|\mathbb{Z}_9| = 9,$$

the possible subgroup sizes are the divisors of 9:

$$1, 3, 9.$$

Size 1

The subgroup of size 1 contains only the identity:

$$\boxed{\{0\}}.$$

Size 3

Start with 3 and keep adding 3 modulo 9:

$$3,$$

$$3 + 3 = 6,$$

$$6 + 3 = 9 \equiv 0 \pmod{9}.$$

So

$$\boxed{\langle 3 \rangle = \{0, 3, 6\}}.$$

Size 9

Start with 1. It generates every element of $\mathbb{Z}_9$:

$$1, 2, 3, 4, 5, 6, 7, 8, 0.$$

So

$$\boxed{\langle 1 \rangle = \mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}}.$$

Therefore, all subgroups of $\mathbb{Z}_9$ are

$$\{0\}, \quad \{0, 3, 6\}, \quad \{0, 1, 2, 3, 4, 5, 6, 7, 8\}.$$

Part B: Subgroups of $\mathbb{Z}_{13}^*$

First calculate the group:

$$\mathbb{Z}_{13}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Since 13 is prime, every number from 1 to 12 is relatively prime to 13.

So

$$|\mathbb{Z}_{13}^*| = 12.$$

By Lagrange's theorem, subgroup sizes must divide 12:

$$1, 2, 3, 4, 6, 12.$$

Now use powers of 2 modulo 13:

$$\begin{aligned} 2^1 &\equiv 2, & 2^2 &\equiv 4, & 2^3 &\equiv 8, & 2^4 &\equiv 3, \\ 2^5 &\equiv 6, & 2^6 &\equiv 12, & 2^7 &\equiv 11, & 2^8 &\equiv 9, \\ 2^9 &\equiv 5, & 2^{10} &\equiv 10, & 2^{11} &\equiv 7, & 2^{12} &\equiv 1 \pmod{13}. \end{aligned}$$

Thus

$$\langle 2 \rangle = \mathbb{Z}_{13}^*.$$

So $\mathbb{Z}_{13}^*$ is cyclic, and its subgroups come from powers of the generator 2.

Size 1

$$\{1\}.$$

Size 2

Use $2^6 \equiv 12 \pmod{13}$:

$$12^1 \equiv 12, \quad 12^2 = 144 \equiv 1 \pmod{13}.$$

So

$$\{1, 12\}.$$

Size 3

Use $2^4 \equiv 3 \pmod{13}$:

$$3^1 \equiv 3, \quad 3^2 = 9, \quad 3^3 = 27 \equiv 1 \pmod{13}.$$

So

$$\{1, 3, 9\}.$$

Size 4

Use $2^3 \equiv 8 \pmod{13}$:

$$\begin{aligned} 8^1 &\equiv 8, & 8^2 &= 64 \equiv 12, \\ 8^3 &= 512 \equiv 5, & 8^4 &= 4096 \equiv 1 \pmod{13}. \end{aligned}$$

So

$$\{1, 5, 8, 12\}.$$

Size 6

Use $2^2 \equiv 4 \pmod{13}$:

$$\begin{aligned} 4^1 &\equiv 4, & 4^2 &= 16 \equiv 3, & 4^3 &= 64 \equiv 12, \\ 4^4 &= 256 \equiv 9, & 4^5 &= 1024 \equiv 10, & 4^6 &= 4096 \equiv 1 \pmod{13}. \end{aligned}$$

So

$$\{1, 3, 4, 9, 10, 12\}.$$

Size 12

The whole group is the subgroup of size 12:

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Main idea

subgroup sizes must divide the group size.

For $\mathbb{Z}_9$, the group size is 9, so the possible sizes are 1, 3, 9. For $\mathbb{Z}_{13}^*$, the group size is 12, so the possible sizes are 1, 2, 3, 4, 6, 12.

Question 3

Prove the subgroup test: if $(S, \oplus)$ is a finite group and S' is a nonempty subset of S such that

$$a \oplus b \in S'$$

for all $a, b \in S'$, then $(S', \oplus)$ is a subgroup of $(S, \oplus)$.

Answer 3

We need to show that S' has the group properties using the same operation $\oplus$.

Closure is already given:

$$a, b \in S' \implies a \oplus b \in S'.$$

Associativity is inherited from the bigger group S .

Now take any element $a \in S'$. Since S' is closed,

$$a, \quad a^2, \quad a^3, \quad \dots$$

all belong to S' , where a^k means applying the operation k times.

Because S' is finite, two powers must repeat:

$$a^i = a^j$$

for some $i < j$. Cancel a^i inside the bigger group S :

$$e = a^{j-i}.$$

So the identity element e belongs to S' .

Also,

$$a^{j-i} = e.$$

Thus

$$a \oplus a^{j-i-1} = e,$$

so a^{j-i-1} is the inverse of a . Since S' is closed, this inverse also belongs to S' .

Therefore S' contains the identity, contains inverses, is closed, and is associative. Hence

$$\boxed{S' \text{ is a subgroup of } S.}$$

Question 4

Show that if p is prime and e is a positive integer, then

$$\phi(p^e) = p^{e-1}(p-1).$$

Answer 4

The numbers not relatively prime to p^e are exactly the multiples of p .

There are p^e residues modulo p^e :

$$0, 1, 2, \dots, p^e - 1.$$

The multiples of p among them are

$$0, p, 2p, \dots, (p^{e-1} - 1)p.$$

So there are

$$p^{e-1}$$

multiples of p .

Therefore the number of residues relatively prime to p^e is

$$p^e - p^{e-1}.$$

Factor:

$$p^e - p^{e-1} = p^{e-1}(p-1).$$

Hence

$$\boxed{\phi(p^e) = p^{e-1}(p-1).}$$

Question 5

Show that for any integer $n > 1$ and for any $a \in \mathbb{Z}_n^*$, the function

$$f_a : \mathbb{Z}_n^* \rightarrow \mathbb{Z}_n^*$$

defined by

$$f_a(x) = ax \bmod n$$

is a permutation of $\mathbb{Z}_n^*$.

Answer 5

The function is a permutation because multiplication by an invertible element can be undone. Since

$$a \in \mathbb{Z}_n^*,$$

we have

$$\gcd(a, n) = 1.$$

So a has an inverse modulo n , call it a^{-1} .

First, f_a maps $\mathbb{Z}_n^*$ back into $\mathbb{Z}_n^*$. If $x \in \mathbb{Z}_n^*$, then

$$\gcd(x, n) = 1.$$

Since both a and x are relatively prime to n ,

$$\gcd(ax, n) = 1.$$

So

$$ax \bmod n \in \mathbb{Z}_n^*.$$

Now show the function is one-to-one. Suppose

$$f_a(x) = f_a(y).$$

Then

$$ax \equiv ay \pmod{n}.$$

Multiply both sides by a^{-1} :

$$x \equiv y \pmod{n}.$$

So different inputs cannot give the same output.

Since $\mathbb{Z}_n^*$ is finite, a one-to-one function from $\mathbb{Z}_n^*$ to itself is also onto. Therefore f_a rearranges the elements of $\mathbb{Z}_n^*$, meaning it is a permutation:

f_a is a permutation of $\mathbb{Z}_n^*$.
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